



Semester II Examinations 2019

Exam Codes	1HDS1, 3BS9, 3BMS2, 4BME1, 4BS2, 4BPT2, 4FM2
Exam	Third Science, Fourth Science Fourth Financial Mathematics and Economics Fourth Mathematics and Education
Module	Non-Linear Systems
Module Code	MP491
Paper No.	1
External Examiners	Professor Yibin Fu
Internal Examiners	Dr Martin Meere
Instructions	Marks are awarded for Question 1 in Section A and the TWO best questions in Section B. All questions carry the same weight.
Duration	TWO HOURS
No. of Pages	Four pages
Discipline	Applied Mathematics
Requirements	Release to Library: Yes Statistical Tables: Yes - Log Tables
Release in Exam Venue	Yes
MCQ	No

A Poincaré diagram is available at the end of this paper

Section A - Answer all parts (a)-(d). ([5 Marks]/part)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = x - x^3.$$

Find all equilibrium points and determine their stability.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point $(x, y) = (0, 0)$ and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations given by

$$\begin{aligned} \frac{dx}{dt} &= x^2 - y, \\ \frac{dy}{dt} &= 1 - 2xy. \end{aligned}$$

- i. Show that this system is Hamiltonian.
- ii. Find the Hamiltonian.

- (d) Consider the map

$$x_{n+1} = 4x_n(1 - x_n)$$

for $n = 0, 1, 2, \dots$. Find all fixed points and determine if they are stable or unstable.

Section B - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = ax - 4x^3,$$

where a is a real parameter.

- i. Find all equilibrium points.
- ii. Sketch the bifurcation diagram, and indicate the behaviour of non-equilibrium solutions using appropriate arrows.
- iii. Find the bifurcation point and classify it.

[10 Marks]

- (b) Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= 2x + y + x^2 - xy, \\ \frac{dy}{dt} &= 3\mu x + \mu y - x^2 y.\end{aligned}$$

Find the value of μ for which a Hopf bifurcation occurs at the origin $(x, y) = (0, 0)$. How does the nature of the equilibrium point change at the Hopf bifurcation?

[10 Marks]

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= y - x, \\ \frac{dy}{dt} &= 1 - xy.\end{aligned}$$

- (a) Locate and classify all equilibrium points in the associated phase plane.

[6 Marks]

- (b) Identify, and sketch separately, the isoclines corresponding to $k = -1, 0, 1, \infty$, where

$$\frac{dy}{dx} = k.$$

[10 Marks]

- (c) Use all four isoclines found in (b) to illustrate the phase-plane portrait for this system.

[4 Marks]

4. Consider the one-dimensional map

$$x_{n+1} = \frac{ax_n}{1 + x_n},$$

where $n = 0, 1, 2, \dots$ and a is a parameter.

- (a) Find all fixed points and investigate for what values of a the model only has non-negative fixed points.

[4 Marks]

- (b) Calculate the stability of the fixed points found in (a) when varying $a \in (-\infty, \infty)$.

[7 Marks]

- (c) Sketch the bifurcation diagram for $-3 \leq a \leq 3$.

[5 Marks]

- (d) Use the results in (a)-(c) and determine if x can undergo any bifurcations when $x > 0$. Explain your answer.

[4 Marks]

INFORMATION: Poincaré Diagram

